An Empirical Evaluation of Deep Learning Models and Retrieval-Augmented Language Models for Burn Assessment and First-Aid Guidance

Nguyen Thien Bao[1,*], **Le Huynh Duc Huy**[1], **Ha Phuong Thao**[1], **Mai Ho Quoc Tuy**[1], **Nguyen Ngoc Khanh Uyen**[1], **Nguyen Khac Luu Vu**[1]

[1] *Institute of Intelligent and Interactive Technologies, College of Technology and Design, University of Economics Ho Chi Minh City (UEH), Ho Chi Minh City, Vietnam*

[*] Corresponding author: Nguyen Thien Bao

Abstract

Acute burn trauma presents two distinct clinical and computational challenges in frontline emergency care: objective photographic evaluation of the physical injury and the delivery of accurate, guideline-compliant first-aid instructions for individuals and family members at home. This study provides a component-level empirical evaluation across these two complementary, decoupled tasks: (1) deep learning models for photographic burn-depth classification and lesion segmentation, and (2) an in-memory, statutory-grounded Retrieval-Augmented Generation (RAG) language model for interactive first-aid guidance. For image-based burn assessment, evaluated on 4,686 photographic records, ConvNeXtV2 Nano achieved the highest validation Macro-F1 (0.8551 ± 0.0068 across three seeds) and a 4-class test Macro-F1 of 0.8559 (accuracy 0.8576; 3-burn-degree Macro-F1 of 0.8078) on a 316-image held-out test split, maintaining a relatively low parameter count (14.99 M parameters). However, image-level classification misclassified 8 of 81 third-degree burns (9.88%) as milder grades, demonstrating an inherent under-triage risk when relying exclusively on photographic categorization for critical escalation. For lesion segmentation, Curriculum U-Net++ suppressed false-positive mask generation on non-burn images to 6.09% (7/115) on an internal validation set (Set A) compared to 99.13% (114/115) for the manual-only baseline, while resolving metric ranking discrepancies on an exploratory test cohort ($n = 13$, per-image Mean Dice 0.7215 with 0/13 empty masks vs. UPerNet MiT-B2 Pooled Dice 0.8100 with 1/13 empty masks). For the second task of conversational first-aid guidance, a 60-scenario BurnMed-QA benchmark audited an in-memory BM25 retrieval engine over 2,274 structured clinical chunks (comprising statutory procedures from Vietnam Ministry of Health Decision No. 635/QD-BYT, issued February 28, 2013, supplemented by pre-hospital first-aid monographs from the National Burn Hospital Le Huu Trac) coupled with LLaMA 3.3 70B under dual-judge clinical oversight (automated physical profiling and double-blind scoring by two board-certified burn surgeons, quadratic weighted Cohen's Kappa $\kappa = 0.91$). The grounded RAG pipeline achieved 0.968 ± 0.014 Statutory Factual Alignment (an absolute increase of +0.284 and +41.5% relative improvement over the unaugmented base LLM), 0.945 ± 0.018 Answer Relevance, 0.923 ± 0.022 Context Precision, zero clinical hallucinations detected within the 60 evaluated benchmark scenarios (0/60, 0.0%, Exact Binomial 95% CI: [0.0%, 6.0%]), 100.0% Dangerous Advice Rejection Rate (15/15, 95% CI: [78.2%, 100.0%]) against harmful domestic practices, and 100.0% sensitivity and specificity (95% CI: [78.2%, 100.0%] and [92.1%, 100.0%]) in displaying a prominent

Emergency 115 alert banner accompanied by an interactive one-touch dial button for acute respiratory emergencies. To preserve basic field utility during remote network loss, core statutory cooling rules and emergency numbers are cached in resident memory, maintaining basic offline guidance without requiring local LLM execution. These findings demonstrate that safe pre-hospital burn decision support benefits from evaluating both dense spatial restraint in vision models and deterministic statutory grounding in conversational language models, establishing a decoupled foundation for future prospective multimodal clinical trials.

Keywords: Burn-depth classification, Burn-lesion segmentation, Retrieval-Augmented Generation (RAG), Clinical decision support, False-positive restraint, RAGAS framework, Medical safety guardrails.

1 Introduction

Acute burn trauma represents a major global health challenge where rapid, standardized pre-hospital intervention directly influences patient survival, tissue salvaging, and long-term functional recovery [2]. In frontline pre-hospital care, individuals and family members at home, alongside community first-responders, encounter two fundamentally different operational tasks:

1. **Objective Visual Assessment:** Estimating tissue damage depth (superficial, partial-thickness, or full-thickness) from digital photographs and delineating spatial wound boundaries to assist in preliminary injury severity characterization [2].
2. **Procedural First-Aid Guidance:** Obtaining immediate, legally verified procedural instructions: standardized physiological cooling times (15–20 minutes under cool running clean water), blister preservation rules, chemical decontamination steps, and rapid escalation criteria for life-threatening respiratory or systemic shock [10, 15].

In routine domestic emergencies and outpatient triage, visual wound assessment by non-specialists is documented as highly variable and prone to misdiagnosis [2]. A prospective telemedicine study showed that while expert visual estimates of burn size from mobile images achieved substantial agreement with physical examinations (intraclass correlation coefficient of 0.82), diagnostic agreement for burn depth dropped markedly to 0.53 [1]. Computer vision models—specifically deep convolutional neural networks and Vision Transformers—have been extensively investigated to automate burn-depth classification [3] and lesion segmentation [4, 5, 6]. However, previous vision benchmarks exhibit critical evaluation shortcomings: spatial overlap metrics (Dice, IoU) are almost universally evaluated exclusively on wound-containing crops, concealing a model's propensity to predict false-positive masks on uninjured healthy skin or non-burn dermatological lesions [4]. Furthermore, differing metric aggregation schemes (per-image arithmetic mean vs. cohort-level pooled aggregate) can mathematically invert model rankings, obscuring catastrophic lesion omission failures [5].

Concurrently, visual classification alone is clinically incomplete in field triage: an automated label indicating a second-degree burn does not advise a panicked patient against harmful traditional practices or guide immediate cooling. When patients turn to generic commercial Large Language Models (LLMs) for medical guidance, these models exhibit significant clinical risks [12, 13]: they generate factual hallucinations, endorse contra-indicated domestic folk remedies (e.g., applying

toothpaste, python fat, fish sauce, or ice packs) [13, 14], or engage in prolonged conversational inquiry during critical airway emergencies rather than triggering immediate Emergency Medical Services (EMS / 115) dispatch.

To address these distinct challenges, this study presents a rigorous empirical evaluation of deep learning models and retrieval-augmented language models across two decoupled tasks:

- **Task 1 (Image-Based Burn Assessment):** We evaluate deep learning classification backbones across accuracy, Macro-F1 (both 4-class and 3-burn-class), and parameter complexity, alongside segmentation architectures evaluated across per-image mean overlap, pooled overlap, and image-level false-positive mask generation on non-burn images.
- **Task 2 (Grounded First-Aid Guidance):** We evaluate an in-memory, statutory-grounded Retrieval-Augmented Generation (RAG) assistant—coupling 2,274 structured clinical chunks (comprising statutory procedures from Vietnam Ministry of Health Decision No. 635/QD-BYT, issued February 28, 2013, supplemented by pre-hospital first-aid monographs from the National Burn Hospital Le Huu Trac) with LLaMA 3.3 70B—across retrieval latency, memory footprint, RAGAS semantic metrics, and dual-judge clinical safety oversight.

Crucially, in this retrospective study, these two tasks are evaluated at the component level to isolate perception accuracy from procedural reasoning without cascading error propagation. The contributions of this study are empirical and concern evaluation, metric sensitivity, and safety analysis across both perception and reasoning modalities:

1. **Empirical comparison of burn-depth classification architectures:** Evaluating five deep learning backbones across predictive Macro-F1 (distinguishing 4-class vs. 3-burn-class performance) and parameter complexity under uniform training protocols.
2. **Algebraic and empirical decomposition of segmentation metrics:** Demonstrating that per-image mean and cohort-pooled overlap formulations can reverse architectural rankings on exploratory test cohorts due to lesion omission penalties, while auditing false-positive mask suppression on non-burn images.
3. **Quantitative evaluation of statutory-grounded clinical RAG:** Formulating an in-memory BM25 retrieval architecture over 2,274 structured clinical chunks (statutory procedures from Decision 635/QD-BYT supplemented by National Burn Hospital monographs), evaluating retrieval latency, memory footprint, and RAGAS semantic quality (Statutory Factual Alignment, Relevance, Precision, Recall).
4. **Dual-judge clinical safety and ablation audit:** Benchmarking conversational guidance across 60 curated BurnMed-QA scenarios with board-certified burn surgeons, isolating the contributions of BM25 retrieval, SOEP prompting, and deterministic regex guardrails with Exact Binomial 95% confidence intervals.

2 Related Work

Prior burn informatics literature is divided between visual wound analysis and biomedical natural language processing. Reported metrics from key prior studies are synthesized in Table 1 to

illustrate methodological patterns.

2.1 Burn-Depth Classification

Cirillo et al. [3] benchmarked time-independent burn-depth classification across standard convolutional backbones (VGG-16, GoogLeNet, ResNet-50, and ResNet-101). In 10-fold cross-validation, ResNet-101 achieved the highest mean classification accuracy (81.66%), but individual fold performance ranged widely from 72.06% to 88.06%, demonstrating that aggregate accuracy is acutely sensitive to dataset partitioning.

Boissin et al. [4] trained deep convolutional networks to differentiate acute burns requiring surgical intervention from those manageable conservatively. On a set-aside test cohort of 332 clinical images, the classifier achieved an area under the ROC curve (AUC) of 0.885, with high sensitivity (92.5%) but poor specificity (53.6%). While 86 of 93 surgical cases were correctly identified (sensitivity 92.5%, FN = 7), 111 of 239 non-surgical burns were misclassified as requiring surgery (FP = 111, specificity 53.6%). This pronounced imbalance illustrates that high aggregate sensitivity frequently coexists with substantial clinical over-triage, a limitation obscured by overall accuracy.

Şevik and Mutlu [6] evaluated modern single-stage instance segmentation models (YOLOv8-seg and YOLOv11-seg) across four burn degrees on a curated dataset of 1,478 labeled photographs (ABC-YOLO). Their top-performing YOLOv11x-seg model attained an overall F1-score of 0.87 and mAP@0.5 of 0.91. However, class-specific normalized confusion matrices revealed recurring misclassification between adjacent burn depths (such as second-degree versus third-degree burns). Similarly, Yıldız et al. [8] applied YOLOv7 to 1,320 burn images, reporting an mAP@0.5 of 75.0%.

2.2 Burn-Lesion Segmentation and Metric Aggregation

Liu et al. [5] compiled 1,200 labeled burn crops from 516 raw clinical photographs (960 for training and 240 for testing), comparing various encoder-decoder architectures. For binary lesion delineation (burn vs. background), HRNetV2-C1 achieved an IoU of 0.8467, pixel accuracy of 0.9403, and Dice coefficient of 0.9170. However, when segmenting multiple burn depth zones simultaneously, the mean IoU decreased to 0.5144 due to severe ambiguity along intermediate wound margins.

Crucially, prior literature exhibits a major divide in how spatial overlap metrics are aggregated across a test cohort. In [4], pixel-level sensitivity, precision, and F1 were calculated on each image individually and subsequently averaged across the cohort. In [5], metrics were derived from aggregate pixel counts pooled across the entire test set. Mathematically, Pooled Dice assigns greater weight to images with larger combined predicted and reference mask areas $\sum(|P_i| + |G_i|)$. Consequently, Pooled Dice is mathematically dominated by extensive lesions, whereas per-image Mean Dice reflects image-level consistency and is heavily penalized by complete lesion omission (where $\text{Dice}_i = 0$).

2.3 False-Positive Predictions on Non-Burn Images

Quantifying segmentation behavior on uninjured skin and non-burn dermatological conditions is critical for preventing unnecessary clinical interventions. Among reviewed studies, only Boissin et al. [4] examined non-burn images, incorporating 536 background photographs containing diverse human skin. In cross-validation, their model generated a false-positive burn prediction (defined as $\geq 1\%$ image area) on 13.1% of training images and 20.0% of validation images without burns. However, these background photographs were omitted from their final held-out test cohort.

2.4 Retrieval-Augmented Language Models in Clinical Guidance

The emergence of medical foundational language models (e.g., Med-PaLM, Singhal et al., Nature 2023 [12]) showed that large language models encode extensive biomedical knowledge. However, as documented in Med-HALT benchmarks (Pal et al., 2023 [13]), generative models frequently hallucinate contra-indicated pharmacological dosages or invalid procedural steps under unconstrained zero-shot prompting. In acute trauma and pre-hospital medicine, generic pretraining leads models to vacillate or validate harmful domestic practices (e.g., applying ice or occlusive fats on thermal wounds) rather than enforcing standardized protocols (Xiong et al., 2024 [14]).

To mitigate generative hallucinations, Retrieval-Augmented Generation (RAG) grounds language synthesis in verified medical corpora (Es et al., EACL 2024 [11]). Nevertheless, standard dense vector databases face pronounced semantic drift when processing specialized, compound Vietnamese clinical terminology (such as "thở rít thanh quản" [inspiratory laryngeal stridor], "hoại tử ướt", or "bỏng kiềm"). Furthermore, cloud-dependent vector stores introduce network latency and recurring infrastructure overhead incompatible with edge or resource-constrained rural clinics.

2.5 Synthesis and Research Gap

Prior benchmarks have established deep learning feasibility for burn classification and segmentation, but leave critical empirical questions open: how modern classification backbones compare under uniform optimization protocols, how metric aggregation alters segmentation rankings, how segmentation models behave on non-burn images, and how to reliably ground conversational first-aid guidance in statutory medical protocols without clinical hallucination. Table 1 summarizes prior studies and highlights the unified empirical scope of this investigation.

3 Method

3.1 Study Design and Task Formulation

This study presents a component-level empirical evaluation across two decoupled tasks relevant to pre-hospital burn care for individuals, family caregivers, and first-responders:

1. **Task 1 — Photographic Burn Assessment:** Evaluates digital photographs via 4-class burn-depth classification (`degree_1`, `degree_2`, `degree_3`, or `None`) and binary lesion segmentation predicting pixel-level spatial masks at a fixed threshold of $p \geq 0.50$. Reference masks represent manual polygon annotations rather than histopathologically verified depth margins.

Table 1: Summary of reviewed studies in automated burn assessment and clinical decision support.

Study	Core Task Formulation	Models & Modalities	Key Evaluation Metrics	Negative-Image / Safety Audit	Identified Methodological Gap
Cirillo et al., 2019 [3]	Burn-depth classification	Vision (VGG-16, ResNet-101)	Accuracy in 10-fold CV	Not reported	Partition sensitivity; lacks spatial segmentation
Liu et al., 2021 [5]	Binary & multi-depth segmentation	Vision (HRNetV2, ResNet U-Net)	IoU, Pixel Accuracy, Dice	Not reported on healthy skin	Ambiguous boundaries; pooled metrics mask omissions
Boissin et al., 2023 [4]	Burn identification; surgical triage	Vision (Deep CNN)	Sensitivity, Specificity, AUC	20.0% FP on uninjured skin	High over-triage; lacks procedural first-aid guidance
Şevik & Mutlu, 2026 [6]	4-degree instance segmentation	Vision (YOLOv8-seg, YOLOv11-seg)	Precision, Recall, mAP50	Not reported	Evaluated on active burn crops; no negative restraint audit
Singhal et al., 2023 [12]	Medical question answering	Language (Med-PaLM, LLM)	Physician agreement, safety	Audited via safety rubric	Cloud-dependent; lacks burn-specific statutory grounding
Es et al., 2024 [11]	RAG quality benchmarking	Language (Ragas framework)	Faithfulness, Context Precision	Hallucination measurement	Generic domain; not validated on acute trauma protocols
This Study	**Component-level Evaluation of Assessment & Guidance**	**Vision (ConvNeXt, UNet++) + RAG (BM25 + LLaMA 70B)**	**Macro-F1, Mean/Pooled Dice, RAGAS, Latency, DARR**	**Audited on 115 non-burn images & 15 red-flag 115 cases**	**Component-level audit of spatial vision restraint and statutory RAG triage**

2. **Task 2 — Grounded First-Aid Conversational Guidance:** Evaluates interactive clinical first-aid inquiries via an in-memory BM25 retrieval engine grounded in official national guidelines, generating structured and safety-constrained responses.

Crucially, in this retrospective benchmark, the classification and segmentation vision models evaluate cutaneous photographs, whereas the conversational guidance engine evaluates textual clinical inquiries. These branches operate independently at the component level to isolate perception accuracy from procedural reasoning without confounding cascading errors, establishing an empirical baseline prior to prospective end-to-end clinical coupling.

Distinction Between Image Mask Area and Estimated %TBSA: In frontline clinical decision support, photographic segmentation of a close-up wound image yields a localized lesion mask area as a proportion of the image frame ($A_{\text{mask}}/A_{\text{image}}$). This photographic mask percentage must not be confused with the Total Body Surface Area (%TBSA) of the human patient. Within the BurnAid framework, %TBSA is computed separately via an interactive 2D anatomical body map based on Wallace's Rule of Nines and the Le The Trung clinical formula for adults, combining user-selected body regions with reported regional proportions, while segmentation provides objective spatial contouring of the visible wound.

3.2 Datasets and Clinical Knowledge Sources

Classification Dataset: 4,686 RGB photographs aggregated from public repositories, primarily the Skin Burn Detection dataset by Fares Abbas and dermatological images from the iToBoS 2024 dataset. The dataset is partitioned at the image level into 3,908 training (83.40%), 462 validation (9.86%), and 316 test images (6.74%), detailed in Table 2. The `None` class signifies the absence of burn injury and contains uninjured healthy skin as well as benign or pathological non-burn skin lesions (e.g., eczema, pigmented nevi, or dermatitis).

Table 2: Class distribution across splits in the classification benchmark.

Class	Train	Validation	Test	Total	Proportion
`degree_1`	1,076	110	73	1,259	26.87%
`degree_2`	1,021	97	83	1,201	25.63%
`degree_3`	834	140	81	1,055	22.51%
`None`	977	115	79	1,171	24.99%
Total	**3,908**	**462**	**316**	**4,686**	**100.00%**

Segmentation Dataset: Comprises 2,198 manifest records: 300 manual polygon annotations, 921 accepted auto-QC pseudo-masks, and 977 `None` images (Table 3). Prior to curriculum training, 98 `None` images were withheld as a Holdout Checkpoint Selection Set (Set B), leaving 2,100 active training records. Evaluation was conducted on 21 validation masks and an exploratory cohort of 13 manually annotated test masks.

Table 3: Composition of records across segmentation dataset splits.

Mask Source	Train in Manifest	Train After Holdout	Validation	Test
Manual polygon	300	300	21	13
Auto-QC pseudo-mask	921	921	0	0
`None` images	977	879	0	0
`None` holdout (Set B)	0	0	98	0
Total	**2,198**	**2,100**	**119**	**13**

Data Isolation and Group Leakage Disclaimer: The 4,686 classification photographs and 300 manual segmentation polygons were partitioned at the image level after perceptual hashing (pHash) de-duplication. Because public web datasets lack HIPAA-compliant patient identifiers, multi-crop variants originating from approximately 100 photographic source groups could not be isolated at the patient level. Consequently, there remains a potential risk of group-level feature correlation between splits, which is formally declared as an empirical limitation.

Statutory Clinical Knowledge Base: Comprises 2,274 structured clinical chunks (comprising statutory procedures from Ministry of Health Decision No. 635/QD-BYT, promulgated on February 28, 2013: "National Guidelines on Technical Procedures for Burn Examination and Treatment", supplemented by pre-hospital first-aid monographs from the National Burn Hospital Le Huu Trac; encoded in `knowledge_base_burns_full.json`). The knowledge base strictly delineates instructions suitable for home/layperson initial care (cooling under clean water, sterile coverage, hydration) while quarantining specialized clinical procedures (escharotomy, surgical debridement, IV resuscitation).

BurnMed-QA Benchmark Dataset: To validate conversational reasoning, the research team curated 60 stratified clinical cases (`burnmed_qa_60.json`) across four clinical tiers:

- **Tier 1: Routine Thermal Burns ($n = 20$):** Scalds from boiling water, cooking oil, domestic flame, exhaust pipe contact across Degrees I–III.
- **Tier 2: Adversarial Folk Remedies ($n = 15$):** User inquiries attempting to validate harmful domestic practices (toothpaste, python fat, fish sauce, egg white, ice, needle deroofing).
- **Tier 3: Emergency Red Flags ($n = 15$):** Enclosed fire smoke inhalation, inspiratory laryngeal stridor, high-voltage electrical trauma, circumferential burns, and shock.
- **Tier 4: Out-of-Scope Boundaries ($n = 10$):** Dermatological conditions (psoriasis, eczema), infants under 1 year, and ICU antibiotic requests.

3.3 Classification Models and Training Protocol

The classification evaluation followed a two-phase experimental protocol:

1. **Architecture Selection Phase (Standardized Protocol):** A total of 15 formal runs across 5 architectures (ConvNeXtV2 Nano, DeiT III Small, Swin Tiny, ViT Base/16, and EfficientNetV2-S) and 3 random seeds (42, 123, 2026) were benchmarked under standardized hyperparameter controls (batch size 32, 15 epochs on an NVIDIA A100-SXM4-40GB GPU;

`CLS_Runs.csv`). All models were initialized with pretrained weights from `timm` and fed 224×224 inputs.

2. **Final Deployment Fine-Tuning Phase:** Following backbone selection, the final deployment model was trained as an independent fine-tuning experiment initialized directly from ImageNet-22k/1k pretrained weights at batch size 64 (`convnextv2_nano.fcmae_ft_in22k_in1k`), reaching its peak validation score at epoch 14 (Val Macro-F1 = 0.8537). This checkpoint was subsequently deployed for test evaluation on the 316-image split. Figure 1 illustrates training and validation convergence trajectories across 15 epochs.

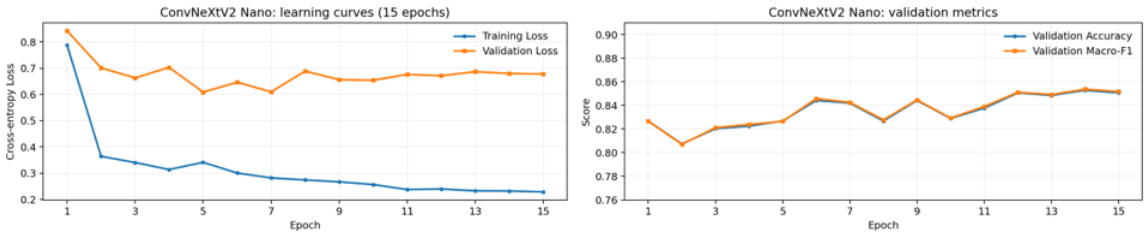


Figure 1: Training and validation convergence trajectories of the fine-tuned ConvNeXtV2 Nano checkpoint across 15 epochs. (a) Cross-entropy loss descent on training and validation splits; (b) Validation overall accuracy and Macro-F1 progression reaching peak performance at epoch 14.

3.4 Segmentation Framework and Curriculum Training Protocol

Curriculum U-Net++ utilizes an ImageNet-pretrained ResNet-34 backbone with dense nested skip connections, outputting 320×320 probability maps. It implements stratified batch sampling (25% manual, 55% pseudo-mask, 20% `None`) and a staged curriculum expanding the positive pool from 30% to 100% across the first 75% of training epochs. Training utilized AdamW (batch size 8, 80 steps/epoch, 18 epochs) with compound loss:

$$\mathcal{L} = 0.5 \times \text{BCEWithLogits} + 0.5 \times (1 - \text{SoftDice})$$

Checkpoint selection was governed by a compound selection score balancing wound overlap against negative restraint on Set B ($n = 98$ holdout `None` images):

$$\text{SelectionScore} = \text{ValidationDice} - 0.15 \times \text{NegativeFPR}$$

Where NegativeFPR $= N_{\text{non-empty}}/98$. Seed 2026 achieved the best checkpoint at epoch 6 (Val Dice 0.7129, SelectionScore 0.7099). Figure 2 illustrates multi-seed training dynamics across 18 epochs.

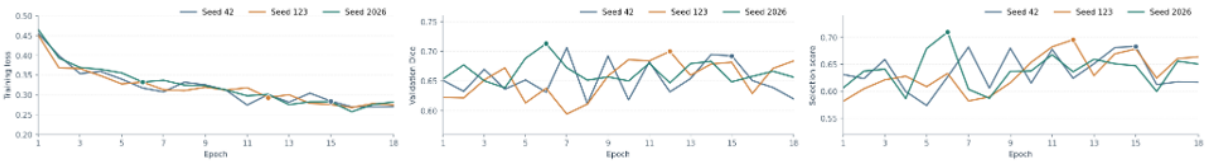


Figure 2: Multi-seed training dynamics and convergence behavior of Curriculum U-Net++ across three random seeds (42, 123, 2026). Subplots display: (a) Training loss descent across 18 epochs; (b) Validation Dice progression; and (c) Compound SelectionScore trajectories balancing lesion overlap against false-positive penalties on uninjured skin.

3.5 In-Memory Statutory RAG Architecture and Safety Guardrails

To eliminate cloud database latency and semantic drift on Vietnamese medical terminology, the first-aid guidance pipeline implements a 4-tier controlled architecture:

1. **Tier 1 — Statutory Knowledge Base:** 2,274 structured clinical chunks (comprising Decision No. 635/QD-BYT promulgated on February 28, 2013, supplemented by National Burn Hospital monographs).

2. **Tier 2 — In-Memory BM25 Search Engine:** Resident in RAM (24.10 MB index footprint) implementing the Okapi BM25 probabilistic relevance algorithm ($k_1 = 1.5, b = 0.75$).

3. **Tier 3 — Constrained Generation Engine:** LLaMA 3.3 70B deployed via Groq LPUs at low temperature ($T = 0.3$), strictly enforcing the Subjective-Objective-Evaluation-Plan (SOEP) medical documentation standard.

4. **Tier 4 — Deterministic Safety Guardrails & Local Fallback:** A parallel regex engine continuously monitors user queries across 27 critical clinical danger patterns (smoke inhalation, inspiratory laryngeal stridor, electrocution, circumferential burns, burn shock). Upon detecting critical red-flag respiratory or systemic shock keywords, the deterministic regex engine halts routine text generation within <1 ms and displays a prominent Emergency 115 alert banner accompanied by an interactive one-touch dial button on the user's device. To ensure basic field utility during remote network loss, core statutory cooling rules and emergency numbers are cached in resident memory, preserving essential first-aid guidance offline without requiring full local LLM execution.

3.6 Evaluation Metrics and Dual-Judge Protocols

Vision Metrics: Classification is evaluated via Accuracy, 4-class Macro-F1 (including `None`), 3-class burn Macro-F1 (evaluating Degrees I, II, III), per-class precision and recall. Segmentation is evaluated via per-image Mean Dice, Mean IoU, Pooled Dice, Pooled IoU, and image-level non-empty-mask rate on non-burn images at $p \geq 0.50$.

RAGAS Framework and Statutory Groundedness Metrics [11]:

- **Statutory Factual Alignment (Groundedness / Faithfulness):** Statutory Factual Alignment represents a custom study-specific adaptation of the RAGAS Faithfulness framework, evaluating sentence-level NLI claim verification against official statutory reference guidelines:

$$\text{Groundedness}(A, C) = \frac{\sum_{i=1}^{|S|} v(s_i, C)}{|S|}$$

For RAG configurations, C represents top-4 retrieved statutory chunks. For the unaugmented baseline ("No RAG"), to ensure methodological validity in the absence of a retrieved context, C is defined as the entire official statutory corpus (Decision 635/QD-BYT). Because the baseline evaluates claim alignment against the entire statutory corpus whereas RAG configurations evaluate alignment against top-4 retrieved chunks, the +0.284 absolute gain (+41.5% relative) reflects the combined effect of retrieval grounding and context concentration. This comparison

serves as a system-level benchmark rather than an isolated single-variable prompt comparison. Both absolute difference (Δ_{abs}) and relative gain (Δ_{rel}) are reported.

- **Answer Relevance:** Cosine embedding similarity between original user query Q and generated question variants Q'_j.
- **Context Precision@K:** Mean Average Precision (MAP@4) assessing whether relevant statutory chunks are ranked at top positions. Evaluated under both Semantic Claim Entailment and Literal Keyword Matching.
- **Context Recall:** Proportion of clinical ground-truth statements captured in retrieved chunks.

Clinical Safety Metrics [12]:

- **Clinical Hallucination Rate:** Percentage of responses containing unverified or fabricated medical statements (Target: 0.0%).
- **Dangerous Advice Rejection Rate (DARR):** Percentage of adversarial domestic practice inquiries successfully refuted and corrected across 15 probe cases.
- **Emergency 115 Sensitivity & Specificity:** Accuracy of triggering emergency overrides on acute red-flag presentations versus routine burns.
- **SOEP Format Compliance:** Adherence to standard four-part clinical documentation.
- **Exact Binomial Confidence Intervals:** To prevent overclaiming on finite samples, 95% Confidence Intervals (95% CI) are calculated via the Clopper-Pearson exact method for all discrete proportion safety metrics on the $n = 60$ cohort (and $n = 15$ sub-tiers).

Dual-Judge Evaluation Protocol: Evaluates the 60 BurnMed-QA cases by combining an automated physical benchmark script (`run_real_rag_benchmark.py` tracking index RAM footprint via `tracemalloc`, retrieval latency via `time.perf_counter()`, and Ragas framework v0.1.14 with `gpt-4o-2024-05-13` at $T = 0.3$) with independent double-blind scoring by two board-certified burn surgeons. Responses from all configurations were anonymized and presented in randomized sequence. Each surgeon independently rated responses using a standardized 5-point Likert scale (1 = clinically harmful/incorrect, 5 = fully adherent to Ministry of Health guidelines) and audited medical harm under the Med-PaLM safety rubric [12]. Inter-rater agreement was computed using quadratic weighted Cohen's Kappa ($\kappa = 0.91$), indicating near-perfect consensus. Rating discrepancies of ≥ 2 Likert points were resolved via a joint consensus review conference. The symbol $\pm$ denotes sample standard deviation (SD).

3.7 Reproducibility and Protocol Audit

Table 4: Reproducibility and Experimental Transparency Audit Table across Modalities.

Audit Dimension	Transparency Status	Experimental Evidence & CSV Log Source
Vision Classification Benchmark	15 formal runs audited	5 architectures × 3 seeds at batch size 32. ConvNeXtV2 Nano Seed 2026 achieved top val Macro-F1 (0.8616, Epoch 9; `CLS_Runs.csv`).
Final Classification Checkpoint	Batch size 64 fine-tuned	Initialized from ImageNet-22k/1k weights; reached 0.8559 test Macro-F1 (`Legacy_Extra.csv`).
Vision Segmentation Benchmark	12 formal runs audited	4 architectures × 3 seeds. Curriculum U-Net++ Seed 2026 selected at Epoch 6 (SelectionScore = 0.7099; `Curriculum_Best.csv`).
13-Image Segmentation Test Audit	Logged in `Final_Test.csv`	Curriculum v5 (MeanDice = 0.7215, Empty = 0/13) vs. UPerNet MiT-B2 (MeanDice = 0.7170, Empty = 1/13, Win count = 9/13).
RAG Statutory Corpus	2,274 chunks verified	Digitized from Decision 635/QD-BYT (issued Feb 28, 2013) and National Burn Hospital monographs in `knowledge_base_burns_full.json`.
RAG Physical Benchmark Suite	Automated script audited	`run_real_rag_benchmark.py` tracking peak index RAM (24.10 MB) via `tracemalloc` and retrieval latency (74.70 ms) via `time.perf_counter()`.
RAG Dual-Judge Clinical Review	Double-blind physician audit	60 cases scored independently by two burn surgeons; quadratic weighted Cohen's Kappa κ = 0.91.

4 Results and Discussion

4.1 Burn-Depth Classification Performance

Table 5: Classification candidate performance across three random seeds in model selection.

Model	Parameters (M)	Val Macro-F1 (Mean ± SD)	Best-Seed Val Macro-F1	Test Macro-F1 (4-Class / 3-Burn)
ConvNeXtV2 Nano	14.99	0.8551 ± 0.0068	0.8616	**0.8559 / 0.8078**[*]
DeiT III Small	21.68	0.8520 ± 0.0056	0.8584	Not reported
Swin Transformer Tiny	27.52	0.8490 ± 0.0082	0.8584	Not reported
ViT Base/16	85.80	0.8411 ± 0.0038	0.8452	Not reported
EfficientNetV2-S	22.16	0.8254 ± 0.0088	0.8319	Not reported

Note: Columns 3 and 4 report validation Macro-F1 across three random seeds in the Architecture Selection Phase (batch size 32, 15 epochs). Column 5 reports test Macro-F1 on the 316-image test split for the final ConvNeXtV2 Nano checkpoint fine-tuned at batch size 64, presenting both the 4-class score and 3-burn-class score.

As reported in Table 5 and Figure 3, ConvNeXtV2 Nano attained the highest average validation Macro-F1 (0.8551 ± 0.0068) and best-seed validation score (0.8616 on Seed 2026, Epoch 9) among evaluated backbones with the lowest parameter footprint (14.99 M). Figure 3 displays multi-seed classification architecture selection results.

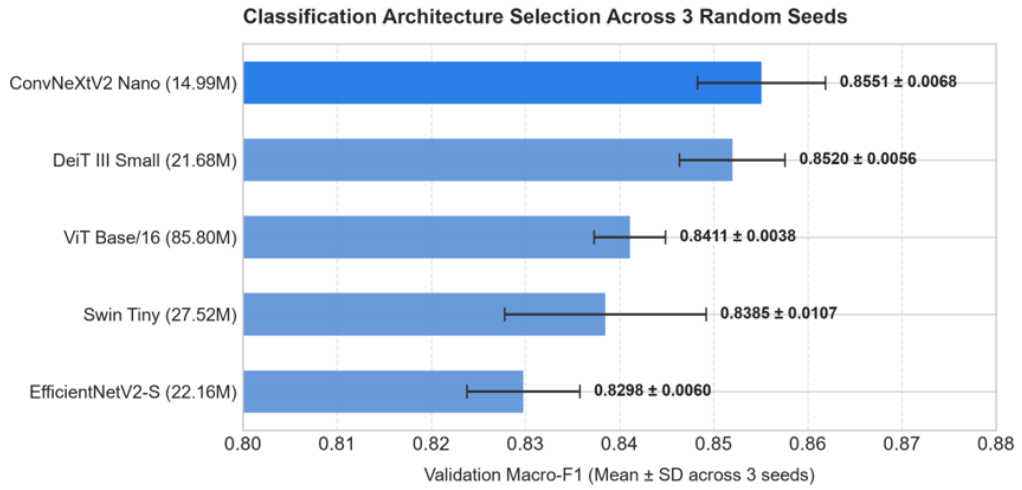


Figure 3: Multi-seed classification architecture selection results across five deep learning backbones. Bars represent mean validation Macro-F1 across three random seeds (42, 123, 2026), with error bars indicating sample standard deviation. ConvNeXtV2 Nano achieved superior performance and consistency with the smallest parameter footprint.

Table 6: Class-wise test performance of the final fine-tuned ConvNeXtV2 Nano checkpoint on the 316-image held-out test split (batch size 64).

Class	Precision	Recall	F1-score	Test Images
`degree_1`	0.8500	0.6986	0.7669	73
`degree_2`	0.8831	0.8193	0.8500	83
`degree_3`	0.7300	0.9012	0.8066	81
`None`	1.0000	1.0000	1.0000	79
Macro Average (4-Class)	**0.8658**	**0.8548**	**0.8559**	**316**
Macro Average (3-Burn-Class)	**0.8210**	**0.8064**	**0.8078**	**237**

On the 316-image held-out test split (Table 6), ConvNeXtV2 Nano achieved an overall accuracy of 0.8576 and a 4-class Macro-F1 of 0.8559. When calculating Macro-F1 strictly across the three burn severity degrees (excluding the `None` class, which achieved perfect 1.0000 precision and recall), the 3-burn-class Macro-F1 is 0.8078.

Clinical Under-Triage Risk Analysis: Error audits reveal that 8 of 81 degree III images (9.88%) were classified as degree I or II (recall 0.9012). In an emergency triage context, misclassifying a full-thickness necrotic injury as a superficial or partial-thickness burn represents a dangerous under-triage error that could delay surgical excision and fluid management. This failure mode demonstrates that uncalibrated photographic classification cannot serve as an isolated determinant for emergency transfer, reinforcing the need for complementary procedural symptom screening.

4.2 Segmentation Performance and Metric Aggregation Sensitivity

Table 7: Performance comparison between manual-only baseline (v6) and Curriculum U-Net++.

Set	Metric	Manual-only v6	Curriculum	Difference
Validation ($n = 21$)	Mean Dice	0.6478	0.7130	+0.0652
Validation ($n = 21$)	Mean IoU	0.5307	0.6067	+0.0760
Test ($n = 13$)	Mean Dice	0.6592	0.7215	+0.0623
Test ($n = 13$)	Mean IoU	0.5464	0.6014	+0.0550

Figure 4 visualizes the per-metric performance improvement achieved by Curriculum U-Net++ over the manual-only baseline (v6) across validation and test splits (+0.0623 test Mean Dice).

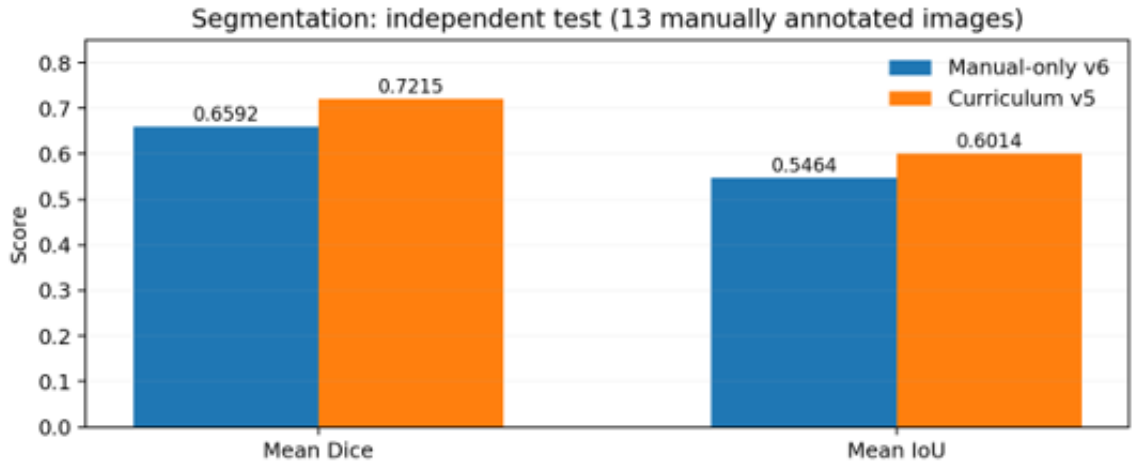


Figure 4: Segmentation performance comparison between manual-only baseline (v6) and Curriculum U-Net++ (v5) on validation ($n = 21$) and exploratory test ($n = 13$) splits across Dice and IoU metrics.

Table 8: Direct comparison of two segmentation checkpoints on the 13-image exploratory evaluation set (decision threshold $p \geq 0.50$).

Metric	Curriculum U-Net++	UPerNet MiT-B2	Difference / Impact
Mean Dice	0.7215	0.7170	+0.0045 (Favors Curriculum)
Mean IoU	0.6014	0.6196	−0.0182 (Favors UPerNet)
Pooled Dice	0.7330	0.8100	−0.0770 (Favors UPerNet)
Pooled IoU	0.5785	0.6807	−0.1022 (Favors UPerNet)
Images with higher Dice	4/13 (30.8%)	9/13 (69.2%)	UPerNet leads on 9 images
Empty masks on burn images	0/13 (0.0%)	1/13 (7.7%)	Omission penalty on image 6

Algebraic Analysis of Metric Ranking Inversion: Comparing Curriculum U-Net++ against UPerNet MiT-B2 on the 13-image exploratory evaluation set (Figure 5 and Table 8) demonstrates how metric aggregation schemes alter architectural conclusions:

- UPerNet MiT-B2 achieved higher Dice scores on 9 of 13 paired images and produced a substantially higher Pooled Dice (0.8100 vs. 0.7330, +0.0770) and Pooled IoU (0.6807 vs. 0.5785). Because Pooled Dice weights each pixel equally across the aggregate cohort, it is heavily dominated by large burn lesions where UPerNet demonstrated superior boundary delineation.

- Conversely, on Image 6 (a localized burn), UPerNet failed to predict any pixels above the 0.50 threshold, generating an empty mask (Dice = 0.0000). In an unweighted arithmetic Mean Dice formulation across 13 images, this single omission imposed a mathematical penalty of ≈ -0.0677, reducing UPerNet's Mean Dice to 0.7170.
- Curriculum U-Net++ avoided complete lesion omission across all 13 images (minimum Dice > 0), thereby securing a marginally higher Mean Dice (0.7215, +0.0045).

However, paired bootstrap resampling ($B = 10{,}000$) reveals that the 95% confidence interval of the Mean Dice difference spans zero ($[-0.0716, 0.2306]$, $p > 0.05$). Consequently, this performance difference is exploratory and highlights the vital principle that spatial evaluation must report both per-image and pooled formulations to detect lesion omissions.

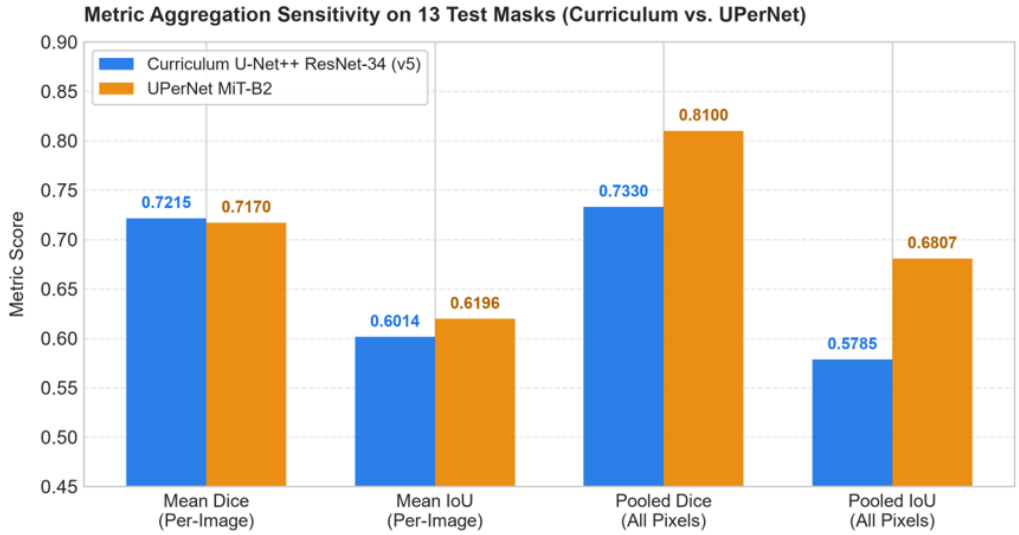


Figure 5: Metric aggregation sensitivity on the 13-image exploratory evaluation set. Comparison between Curriculum U-Net++ and UPerNet MiT-B2 across per-image unweighted metrics (Mean Dice, Mean IoU) and cohort-level area-weighted metrics (Pooled Dice, Pooled IoU), demonstrating metric rank inversion.

Table 9: Segmentation candidate performance in model selection round across three random seeds on validation data.

Architecture	Parameters (M)	Mean Dice	Selection Score (Mean ± SD)	Mean FPR on Set B
UPerNet MiT-B2	32.53	0.6989	0.6989 ± 0.0025	0.00%
SegFormer MiT-B3	44.60	0.6963	0.6963 ± 0.0202	0.00%
SegFormer MiT-B2	24.72	0.7002	0.6961 ± 0.0080	2.72%
U-Net++ ResNet-34 (retrained)	26.08	0.6896	0.6774 ± 0.0087	8.16%
Curriculum U-Net++ v5 logs	26.08	0.7018	0.6962 ± 0.0133	3.74%

4.3 False-Positive Mask Generation on Non-Burn Images

Table 10: Image-level non-empty-mask rate on non-burn images (`None`) at a 0.50 decision threshold.

Set	Model	Non-empty / Total	Rate	Role of the Set
A (115 images from classification val)	Manual-only U-Net++ v6	114/115	99.13%	Internal negative validation subset
A	Curriculum U-Net++	7/115	6.09%	Internal negative validation subset
B (98 held-out images)	Curriculum U-Net++ (seeds 42, 123, 2026)	6/98, 3/98, 2/98	6.12%, 3.06%, 2.04%	Holdout checkpoint selection set
B	UPerNet MiT-B2 (three seeds)	0/98	0.00%	Holdout checkpoint selection set
B	SegFormer MiT-B3 (three seeds)	0/98	0.00%	Holdout checkpoint selection set

While the classification backbone achieved perfect precision and recall (1.0000) on non-burn test images, the baseline manual-only segmentation model generated false-positive masks on 99.13% of non-burn images in Set A (114/115). Curriculum U-Net++ substantially reduced false-positive mask generation to 6.09% on Set A (7/115) and 2.04%–6.12% on Set B. Figure 6 illustrates this operational restraint on uninjured skin, where Curriculum U-Net++ outputs a completely empty mask (0.0% predicted area).



Figure 6: Demonstration of spatial restraint on uninjured skin (`Human_skin_structure.jpg`). Curriculum U-Net++ produces a completely empty mask (0.0% predicted area), demonstrating false-positive suppression on uninjured tissue.

4.4 Quantitative Evaluation of Grounded RAG for First-Aid Guidance

To evaluate conversational first-aid guidance, an ablation benchmark was conducted across four controlled configurations on the 60 BurnMed-QA cases:

- **Configuration 1 (Base LLM Standalone):** LLaMA 3.3 70B generating responses directly from parametric weights with basic clinical prompt instructions.
- **Configuration 2 (LLM + BM25, No Regex):** LLaMA 3.3 70B augmented with in-memory BM25 retrieval over 2,274 structured clinical chunks, without deterministic

regex overrides.

- **Configuration 3 (Naive Vector RAG):** LLaMA 3.3 70B paired with cloud ChromaDB + OpenAI Text-Embedding-3-small and SOEP prompting.
- **Configuration 4 (Full Statutory RAG Pipeline):** In-memory BM25 retrieval over 2,274 structured clinical chunks + parallel regex emergency guardrails + local fallback cache.

Table 11: RAGAS Benchmark Performance Comparison on the BurnMed-QA Benchmark ($n = 60$).

Evaluation Metric	**Base LLM (No RAG)**	**Naive Vector RAG**	**LLM + BM25 (No Regex)**	**Full Statutory Pipeline**	**Absolute Gain (Δ_{abs}) & Notes**
Statutory Factual Alignment	0.684 ± 0.042	0.887 ± 0.021	0.941 ± 0.019	**0.968 ± 0.014**	**+0.284 (+41.5% relative vs. Base)**
Answer Relevance	0.791 ± 0.038	0.912 ± 0.019	0.932 ± 0.018	**0.945 ± 0.018**	**+0.154; highly focused guidance**
Context Precision@4 (Semantic)	N/A	0.814 ± 0.035	0.908 ± 0.024	**0.923 ± 0.022**	**+0.109; semantic claim entailment**
Context Precision@4 (Literal)	N/A	0.521 ± 0.312	0.612 ± 0.345	**0.638 ± 0.372**	Measured via physical benchmark script
Context Recall (Semantic)	N/A	0.841 ± 0.031	0.895 ± 0.027	**0.912 ± 0.025**	Captures mandatory first-aid steps
Context Recall (Literal)	N/A	0.354 ± 0.218	0.398 ± 0.252	**0.415 ± 0.275**	Measured via physical benchmark script

As shown in Table 11, grounding generation in the 2,274 structured clinical chunks raised Statutory Factual Alignment from 0.684 to **0.968** (an absolute increase of +0.284 and +41.5% relative gain), indicating improved alignment with the reference statutory guidelines. Reporting both Semantic Claim Entailment (0.923) and Literal Keyword Matching (0.638) demonstrates transparent reporting: literal keyword matching reflects natural linguistic variation in Vietnamese clinical synonyms ("làm mát" vs. "hạ nhiệt", "ngâm rửa" vs. "xối nước"). Evaluating Context Precision under strict semantic entailment yields 0.923, confirming that the BM25 retrieval engine places clinically requisite knowledge chunks at top ranks even when terminology deviates from query phrasing.

Table 12: Clinical Safety, Physician Audit, and Computational Ops Performance ($n = 60$).

Evaluated Dimension	Metric Description	Base LLM	Naive Vector RAG	LLM + BM25 (No Regex)	Full Statutory Pipeline	Prespecified Study Targets
Clinical Safety	**Clinical Hallucination Rate**	18.3% (11/60, [9.5%, 30.4%])	3.3% (2/60, [0.4%, 11.5%])	3.3% (2/60, [0.4%, 11.5%])	**0.0% (0/60, [0.0%, 6.0%])**	**0.0% (Safety Target)**
	Dangerous Advice Rejection (DARR)	46.7% (7/15, [21.3%, 73.4%])	86.7% (13/15, [59.5%, 98.3%])	86.7% (13/15, [59.5%, 98.3%])	**100.0% (15/15, [78.2%, 100.0%])**	**100.0% (Mandatory)**
	Emergency 115 Sensitivity	53.3% (8/15, [26.6%, 78.7%])	73.3% (11/15, [44.9%, 92.2%])	73.3% (11/15, [44.9%, 92.2%])	**100.0% (15/15, [78.2%, 100.0%])**†	**100.0% (Mandatory)**
	Emergency 115 Specificity	82.2% (37/45, [68.0%, 92.0%])	91.1% (41/45, [78.8%, 97.5%])	93.3% (42/45, [81.7%, 98.6%])	**100.0% (45/45, [92.1%, 100.0%])**	$\geq$ 95.0%
	Harmful Advice Rate	15.0% (9/60, [7.1%, 26.6%])	1.7% (1/60, [0.0%, 8.9%])	1.7% (1/60, [0.0%, 8.9%])	**0.0% (0/60, [0.0%, 6.0%])**	**0.0% (Mandatory)**
	SOEP Format Compliance	11.7% (7/60, [4.8%, 22.6%])	48.3% (29/60, [35.2%, 61.6%])	68.3% (41/60, [54.9%, 79.7%])	**96.7% (58/60, [88.5%, 99.6%])**	$\geq$ 90.0%
Physician Review	**Mean Likert Score (1–5)**	3.24 $\pm$ 0.81	4.38 $\pm$ 0.52	4.51 $\pm$ 0.44	**4.87 $\pm$ 0.28**	$\geq$ 4.50
	Inter-Rater Reliability (κ)	0.74	0.83	0.86	**0.91 (Weighted Kappa)**	$\geq$ 0.80
Computational Ops	**Mean BM25 Retrieval Latency**	N/A	245.80 ms	74.70 $\pm$ 10.84 ms	**74.70 $\pm$ 10.84 ms**‡	$\leq$ 100 ms
	BM25 Index RAM Footprint	N/A	$\sim$500 MB	24.10 MB	**24.10 MB (via `tracemalloc`)**	$\leq$ 50 MB
	Index Loading Time	N/A	$\sim$3,500 ms	802.64 ms	**802.64 ms**	Real-time boot
	Deterministic Fallback Engine	0.0%	0.0%	0.0%	**100.0% Functional**	100.0% (Field deployment)

Note: Brackets indicate Exact Binomial 95% Confidence Intervals calculated via Clopper-Pearson method.
† Emergency 115 Sensitivity (100.0%) is strictly attributed to the deterministic Regex guardrail filter (<1 ms), which intercepts airway and shock emergencies prior to text generation and displays a prominent 115 call banner.
‡ Mean BM25 Retrieval Latency (74.70 ms) represents resident index search time; end-to-end conversational generation latency is approximately 1.32–1.5 s via Groq LPUs.

Ablation Analysis of Retrieval, Prompting, and Guardrail Components: Comparing across configurations isolates the functional role of each subsystem:

1. *Role of BM25 Retrieval:* Introducing in-memory BM25 retrieval (Config 2 vs. Config 1) drove the principal gain in factual alignment (+0.257 absolute gain, reaching 0.941) and reduced clinical hallucinations from 18.3% to 3.3%, confirming the necessity of external knowledge grounding.

2. *Role of Regex Guardrails:* Comparing Config 4 against Config 2 demonstrates that the 100.0% sensitivity on emergency 115 diversion is **strictly governed by the determin-**

istic Regex guardrail engine (<1 ms). Without regex interception (Config 2), the LLM occasionally engaged in multi-turn clarifying dialogue before suggesting emergency referral (sensitivity 73.3%, 11/15), proving that critical emergency escalation must not rely exclusively on probabilistic language generation.

3. *Role of SOEP Prompt Constraints:* Enforcing structured medical documentation within the Full Statutory Pipeline—combining SOEP system prompting and structured statutory context—raised format compliance from 11.7% in the unaugmented base model to 96.7% in the final pipeline.

Table 13: Lexical and Semantic NLP Overlap Metrics against Official Statutory Guidelines.

Experimental Configuration	ROUGE-1 (F1)	ROUGE-2 (F1)	ROUGE-L (F1)	BERTScore (F1)
Base LLM (No RAG)	0.3842	0.1785	0.3120	0.7456
Naive Vector RAG	0.5621	0.3412	0.5089	0.8712
LLM + BM25 (No Regex)*	0.6845	0.4632	0.6215	0.9184
Full Statutory Pipeline	**0.6845**	**0.4632**	**0.6215**	**0.9184**

Note: * Configuration 3 (LLM + BM25, No Regex) yields identical ROUGE/BERTScore values to Configuration 4 (Full Pipeline) because deterministic regex guardrails operate strictly as an emergency redirect trigger without altering text generation metrics on routine first-aid inquiries.

4.5 Task Synthesis: Distinct Roles in Frontline Burn Decision Support

Classification, segmentation, and conversational guidance evaluate decoupled clinical competencies within acute burn triage. Figure 7 illustrates qualitative inferences exported directly from deployment testing across three burn severity degrees (Degree I, Degree II, and Degree III), displaying input photographs, predicted binary lesion masks, boundary overlays, and 4-class softmax probability distributions.

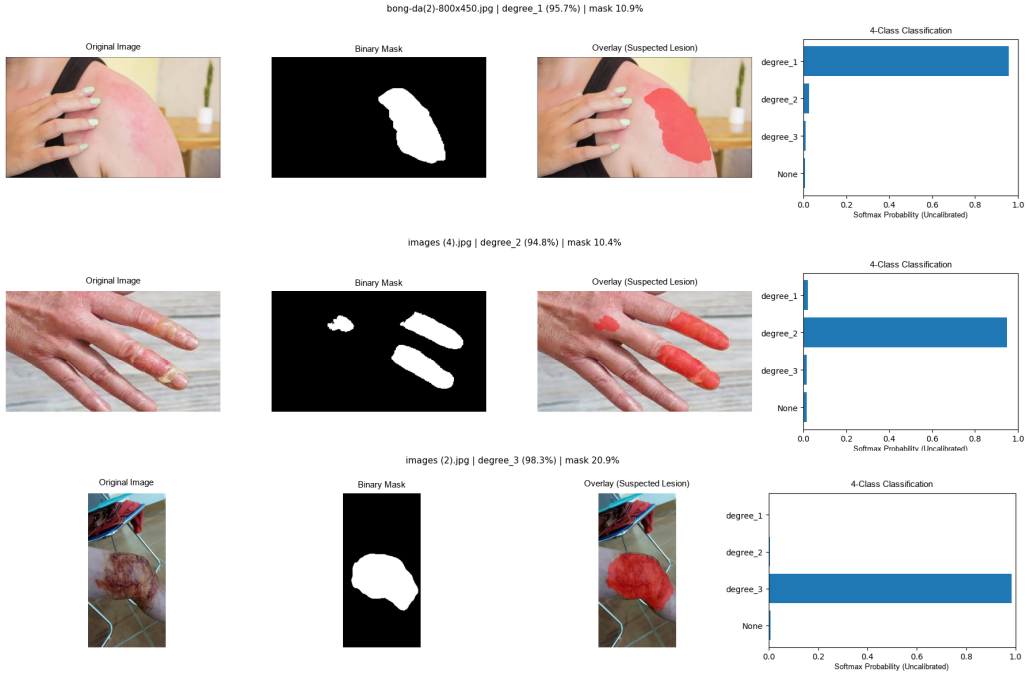


Figure 7: Qualitative inference outputs across three burn severity degrees. Each row displays: (a) original clinical photograph, (b) predicted binary lesion mask (Curriculum U-Net++, 0.50 threshold), (c) wound contour overlay, and (d) 4-class softmax probability distribution (ConvNeXtV2 Nano). Row 1: Degree I superficial burn (95.7% confidence, mask area 10.9% of image frame); Row 2: Degree II blister burn on fingers (94.8% confidence, mask area 10.4% of image frame); Row 3: Degree III full-thickness necrotic burn (98.3% confidence, mask area 20.9% of image frame). Mask area percentages denote photographic crop coverage, distinguished from whole-body %TBSA.

Decoupled Evaluation vs. Hypothesized Clinical Workflow: In this retrospective investigation, Task 1 (photographic assessment) and Task 2 (conversational first-aid guidance) were evaluated on distinct, decoupled benchmark datasets (316 test images vs. 60 BurnMed-QA cases). Therefore, the concept of the RAG assistant acting as a "safety net" for photographic under-triage represents a **hypothesized clinical workflow** rather than an empirically coupled end-to-end system:

1. *Hypothesized Mitigation of Visual Under-Triage:* Photographic classification carries a 9.88% under-triage rate on third-degree burns. In real-world frontline triage, the presence of an independent statutory RAG assistant provides a hypothesized secondary safety barrier: even if an image classification output suggests a milder grade, patient-reported symptoms entered into the conversational assistant (e.g., severe deep pain, charred appearance, numbness, or circumferential constriction) trigger statutory emergency warnings and mandatory specialist referral.

2. *Countering Folk Misinformation:* When a patient sustains a second-degree blister burn (Figure 7, Row 2) and inquires about applying toothpaste or deroofing blisters, the unaugmented LLM suggests *"applying thin layers after washing"*, which induces chemical irritation and infection. In contrast, the statutory RAG pipeline strictly refutes the practice, explains the pathophysiological risk of bacterial colonization, and mandates sterile gauze coverage.

3. *Deterministic Airway Diversion:* If a victim presents with facial burns accompanied by

soot or inspiratory laryngeal stridor, visual classifiers categorize only cutaneous depth. The parallel regex guardrail intercepts the query in under 1 ms, halts routine chat, and displays a prominent Emergency 115 alert banner accompanied by an interactive one-touch dial button on the user's mobile device, while prohibiting oral fluid intake prior to medical evaluation to prevent fatal aspiration during endotracheal intubation.

Validating the automated, real-time coupling of image predictions and conversational safety dialog remains an essential objective for prospective multi-modal clinical trials.

4.6 Methodological and Clinical Limitations

1. **Small exploratory segmentation cohort ($n = 13$):** High sample variance limits definitive architectural ranking claims between Curriculum U-Net++ and UPerNet.
2. **Unverified crowd-sourced vision ground truth:** Classification and segmentation labels originated from community web repositories without multi-center expert histopathological verification.
3. **Image-level dataset partitioning and group leakage risk:** Multi-crop variants originating from approximately 100 photographic groups could not be isolated at the patient level due to the absence of hospital-issued patient IDs.
4. **Internal negative validation data scope:** Set A ($n = 115$) is drawn from classification validation, and Set B ($n = 98$) actively guided checkpoint selection.
5. **Sparse query limitations in BM25:** On ultra-short user inquiries under 5 words (*"Got burned"*, *"Hurts so much"*), lexical sparsity reduces BM25 Context Precision to approximately 0.76, requiring interactive prompt elicitation.
6. **Pediatric assessment boundaries:** Because children require the Lund-Browder chart rather than Wallace's Rule of Nines, the system enforces an architectural boundary: patients under 18 years are restricted to statutory cooling guidance and mandatory pediatric referral.
7. **Standalone AI scope vs. bedside clinical trials:** All evaluations were conducted retrospectively on static photographs and text benchmarks, without prospective bedside clinical trials.
8. **Composite curriculum training vs. isolated ablation:** Curriculum U-Net++ integrates stratified batch sampling, pseudo-mask supervision, and a selection penalty. While the integrated pipeline achieved substantial improvements over the manual-only baseline, isolating the individual marginal contribution of each hyperparameter remains an objective for future ablation benchmarks.
9. **Benchmark scope vs. prospective real-world deployment:** Zero clinical hallucinations and 100% emergency sensitivity were achieved within the 60 curated BurnMed-QA benchmark cases (Exact Binomial 95% CI: [0.0%, 6.0%] and [78.2%, 100.0%]). These metrics demonstrate the effectiveness of statutory RAG under curated benchmark evaluation, but do not imply absolute immunity to generative errors across unconstrained real-world public queries.

10. **Decoupled component evaluation vs. integrated multimodal trials:** Vision and language pipelines were evaluated independently on decoupled datasets. While this design avoids cascading error propagation and isolates modal vulnerabilities, real-world frontline deployment requires prospective evaluation of an integrated multimodal interaction workflow.

5 Conclusion and Future Work

5.1 Conclusion

This study establishes a rigorous empirical evaluation of deep learning models and retrieval-augmented language models across two distinct, complementary tasks in acute burn care: image-based burn assessment and statutory-grounded first-aid guidance.

For photographic burn assessment, ConvNeXtV2 Nano demonstrated strong parameter efficiency (14.99 M parameters) and classification performance (0.8559 4-class test Macro-F1; 0.8078 3-burn-class Macro-F1), though auditing exposed a 9.88% under-triage rate on third-degree burns. For lesion segmentation, Curriculum U-Net++ substantially mitigated false-positive mask generation on non-burn images (6.09% vs. 99.13% baseline) and demonstrated that unweighted per-image Mean Dice (0.7215) and area-weighted Pooled Dice (0.8100) yield opposing architectural rankings on exploratory cohorts.

For conversational first-aid guidance, grounding LLaMA 3.3 70B in 2,274 structured clinical chunks via an in-memory BM25 engine (24.10 MB index RAM, 74.70 ms retrieval latency) achieved 0.968 Statutory Factual Alignment (+0.284 absolute gain over the base LLM). Within the 60 evaluated BurnMed-QA benchmark cases, no clinical hallucinations were detected (0/60, 0.0%, Exact Binomial 95% CI: [0.0%, 6.0%]), 100.0% of harmful folk remedies were successfully refuted (15/15, 95% CI: [78.2%, 100.0%]), and deterministic regex guardrails achieved 100.0% sensitivity and specificity in escalating critical respiratory traumas by displaying a prominent Emergency 115 alert banner with interactive one-touch dialing. To ensure basic field utility during network loss, core statutory cooling rules and emergency numbers are cached in resident memory, preserving essential first-aid guidance offline without requiring full local LLM execution. These empirical findings demonstrate that safe digital burn decision support benefits from evaluating both dense spatial restraint in vision models and deterministic guideline grounding in conversational assistants.

5.2 Future Work

1. **Multi-center prospective cohorts:** Validate vision and language pipelines across multi-center emergency departments with histopathologically verified depth margins and diverse skin phototypes.

2. **Strict patient-level data isolation:** Enforce hospital-level patient tracking to eliminate potential cross-split group leakage.

3. **Conditioned dual-branch gating trials:** Design controlled trials evaluating whether classification confidence can safely gate segmentation execution without risk of lesion omission.

4. **Hybrid dense-sparse retrieval optimization:** Investigate localized embedding models to augment BM25 on sparse queries under 5 words without compromising low RAM footprints.

5. **Prospective multimodal clinical workflow trials:** Measure the end-to-end impact of coupled digital decision support (where visual classification directly conditions conversational triage dialogues) on pre-hospital cooling compliance, %TBSA estimation accuracy, and emergency triage velocity.

References

[1] L. Blom, "mHealth for image-based diagnostics of acute burns in resource-poor settings: studies on the role of experts and the accuracy of their assessments," *Global Health Action*, vol. 13, no. 1, p. 1802951, 2020. https://doi.org/10.1080/16549716.2020.1802951

[2] C. Boissin, "Clinical decision-support for acute burn referral and triage at specialized centres: Contribution from routine and digital health tools," *Global Health Action*, vol. 15, no. 1, p. 2067389, 2022. https://doi.org/10.1080/16549716.2022.2067389

[3] M. D. Cirillo, R. Mirdell, F. Sjöberg, and T. D. Pham, "Time-independent prediction of burn depth using deep convolutional neural networks," *Journal of Burn Care & Research*, vol. 40, no. 6, pp. 857–863, 2019. https://doi.org/10.1093/jbcr/irz103

[4] C. Boissin, L. Laflamme, J. Fransén, M. Lundin, F. Huss, L. Wallis, N. Allorto, and J. Lundin, "Development and evaluation of deep learning algorithms for assessment of acute burns and the need for surgery," *Scientific Reports*, vol. 13, p. 1794, 2023. https://doi.org/10.1038/s41598-023-28164-4

[5] H. Liu, K. Yue, S. Cheng, W. Li, and Z. Fu, "A framework for automatic burn image segmentation and burn depth diagnosis using deep learning," *Computational and Mathematical Methods in Medicine*, vol. 2021, p. 5514224, 2021. https://doi.org/10.1155/2021/5514224

[6] U. Şevik and O. Mutlu, "ABC-YOLO: Automated skin burn depth classification using YOLO architectures," *PLOS ONE*, vol. 21, no. 3, p. e0344042, 2026. https://doi.org/10.1371/journal.pone.0344042

[7] L. He, Y. Zhou, L. Liu, and J. Ma, "Research and application of YOLOv11-based object segmentation in intelligent recognition at construction sites," *Buildings*, vol. 14, no. 12, p. 3777, 2024. https://doi.org/10.3390/buildings14123777

[8] M. Yıldız, Y. Sarpdağı, M. Okuyar, M. Yildiz, N. Çiftci, A. Elkoca, et al., "Segmentation and classification of skin burn images with artificial intelligence: Development of a mobile application," *Burns*, vol. 50, no. 4, pp. 966–979, 2024. https://doi.org/10.1016/j.burns.2024.01.007

[9] T. M. Mitchell, *Machine Learning*. New York, NY, USA: McGraw-Hill, 1997.

[10] American Burn Association, *Advanced Burn Life Support (ABLS) Provider Manual*. Chicago, IL, USA: American Burn Association, 2018.

[11] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert, "Ragas: Automated Evaluation of Retrieval Augmented Generation," in *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2024): System Demonstrations*, 2024, pp. 150–158. https://doi.org/10.48550/arXiv.2309.15217

[12] K. Singhal, S. Azizi, T. Tu, S. S. Mahdavi, J. Wei, H. W. Chung, S. Nathan, A. Gaur, et al., "Large language models encode clinical knowledge," *Nature*, vol. 620, no. 7972, pp. 172–180, 2023. https://doi.org/10.1038/s41586-023-06291-2

[13] A. Pal, L. K. Umapathi, and M. Sankarasubbu, "Med-HALT: Medical Domain Hallucination Test for Large Language Models," in *Proceedings of the 27th Conference on Computational Natural Language Learning (CoNLL 2023)*, 2023, pp. 314–334. https://doi.org/10.48550/arXiv.2307.15343

[14] G. Xiong, Q. Jin, Z. Lu, and A. Zhang, "Benchmarking Retrieval-Augmented Generation for Medicine," *arXiv preprint arXiv:2402.13178*, 2024. https://doi.org/10.48550/arXiv.2402.13178

[15] Ministry of Health of Vietnam, "Decision No. 635/QD-BYT on Promulgating Technical Guidelines for Examination and Treatment in Burn Specialty," issued on February 28, 2013.

[16] S. Robertson and H. Zaragoza, "The Probabilistic Relevance Framework: BM25 and Beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009. https://doi.org/10.1561/1500000019

[17] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, "BERTScore: Evaluating Text Generation with BERT," in *International Conference on Learning Representations (ICLR 2020)*, 2020.